

CUSTOMER CHURN PREDICTION FOR TELECOM RETENTION

**BUILDING A MACHINE LEARNING PIPELINE TO
REDUCE SUBSCRIBER LOSS**

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BUSINESS UNDERSTANDING

The Challenge	Telecom churn leads to significant revenue loss and high acquisition costs
Objective	Develop a classification model to predict which customers are likely to cancel their service.
Stakeholders	Customer Retention, Marketing, and Business Analysts.
Value	Enable proactive engagement (discounts, promotions) before the customer leaves.

SUCCESS METRICS (STRATEGY)

Primary Metric: Recall	<i>Why?</i> It is more expensive to miss a churner (lost revenue) than to send a discount to a loyal customer (small cost).
Precision & F1-Score:	To ensure the marketing budget isn't wasted on too many false alarms.
Accuracy	Used as a secondary baseline.

Understanding the Dataset

CONTENT:

- 3333 CUSTOMERS, 21 FEATURES
- INCLUDES USAGE, CHARGES, AND SERVICE CALLS
- TARGET VARIABLE: CHURN

The heatmap shows relationships between features, helping us understand which variables are related to churn behavior.

DATA PREPARATION WORKFLOW

Cleaning	Dropping unique identifiers (Phone Number).
Encoding	Converting "Yes/No" plans into binary (0 and 1).
Splitting	80/20 Train-Test split to ensure the model
Scaling	Standardizing numerical features so usage minutes and call counts are on the same scale.

Data Visualization

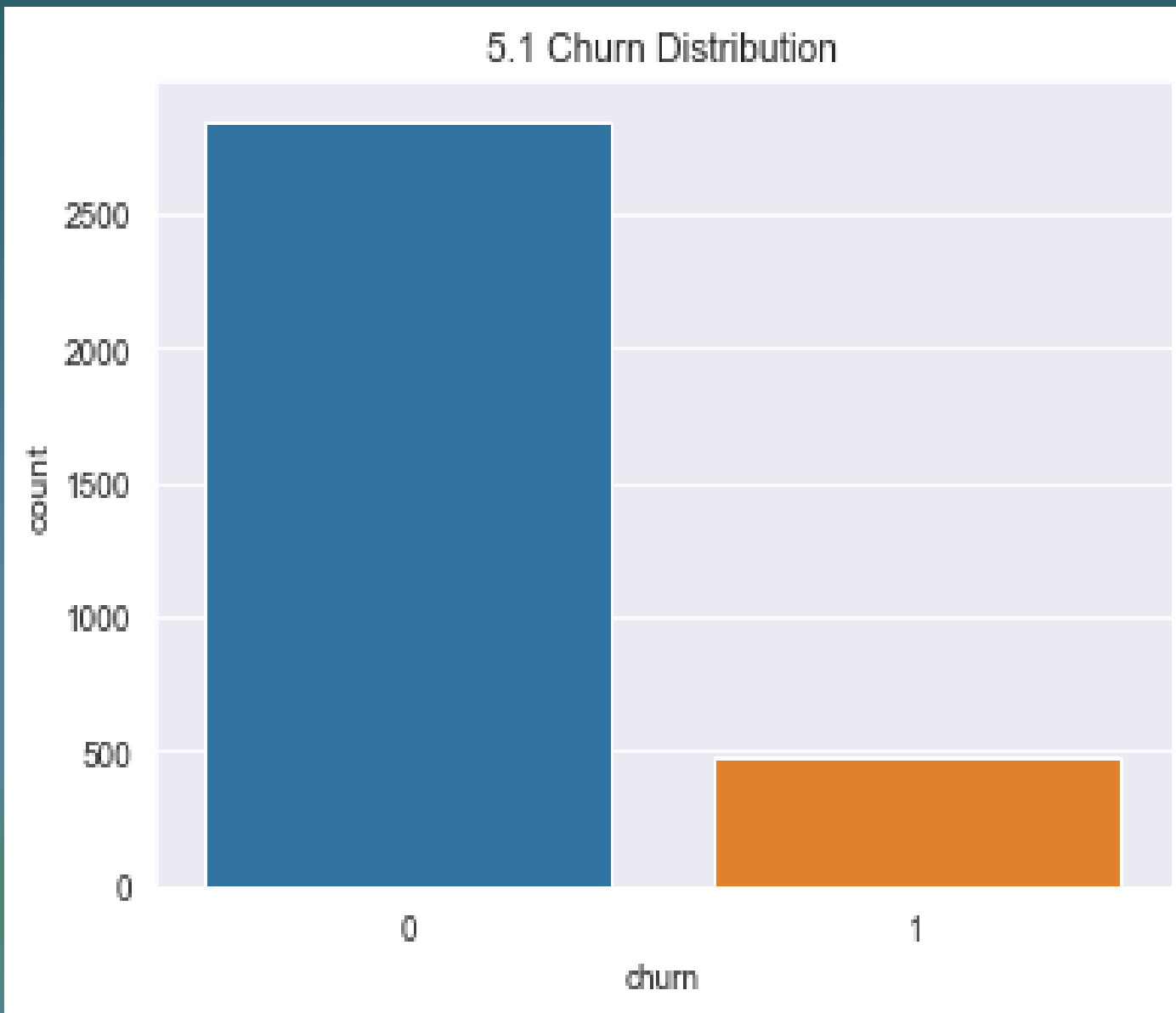
Churn Distribution

Insight

Majority of customers do not churn.

Confirms class imbalance (~85% vs ~15%).

Reinforces why Recall is the key metric.



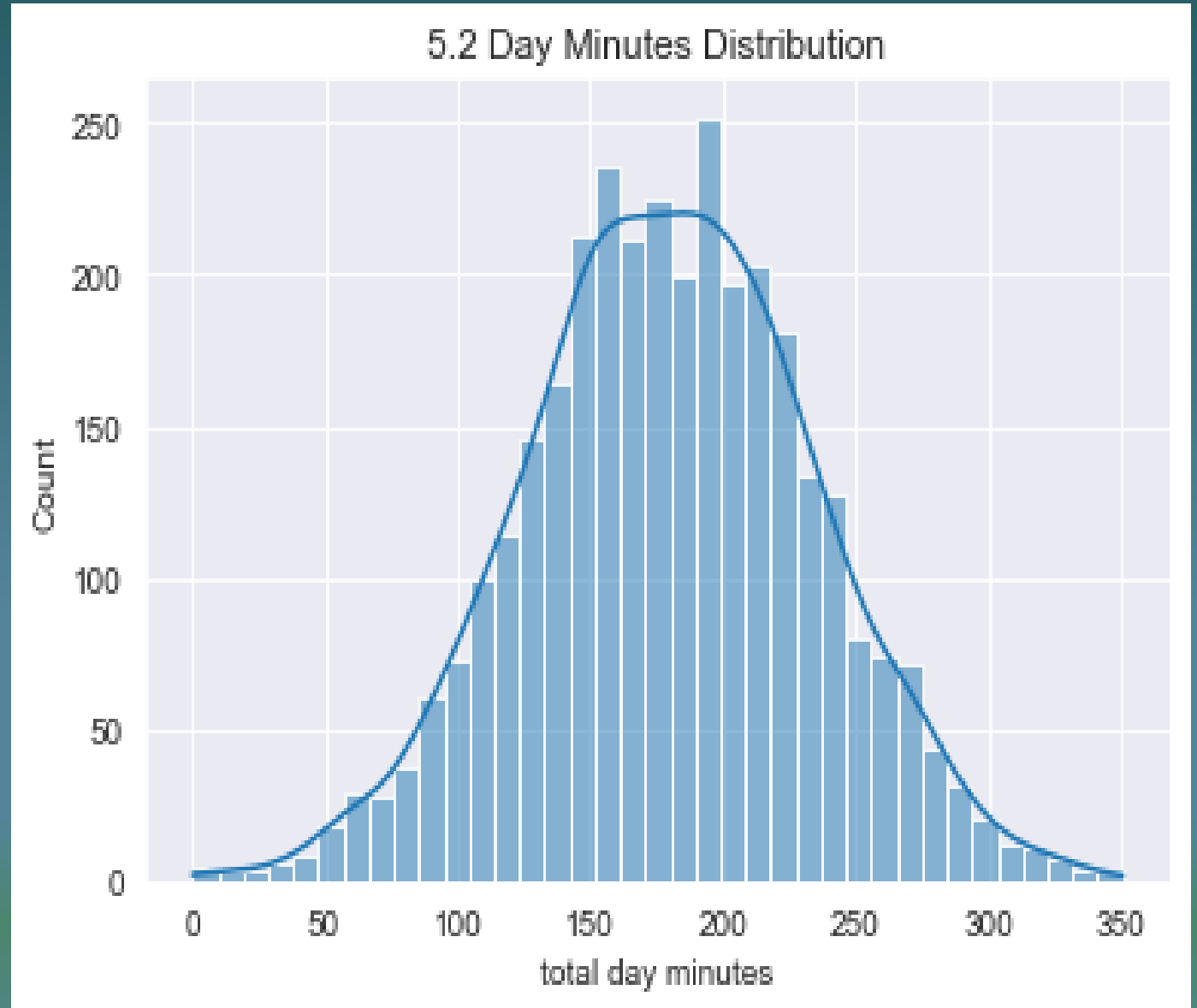
Distribution of Total Day Minutes

Insight

Data is roughly normally distributed.

Most customers have moderate usage.

Few extreme high-usage customers (potential premium users).



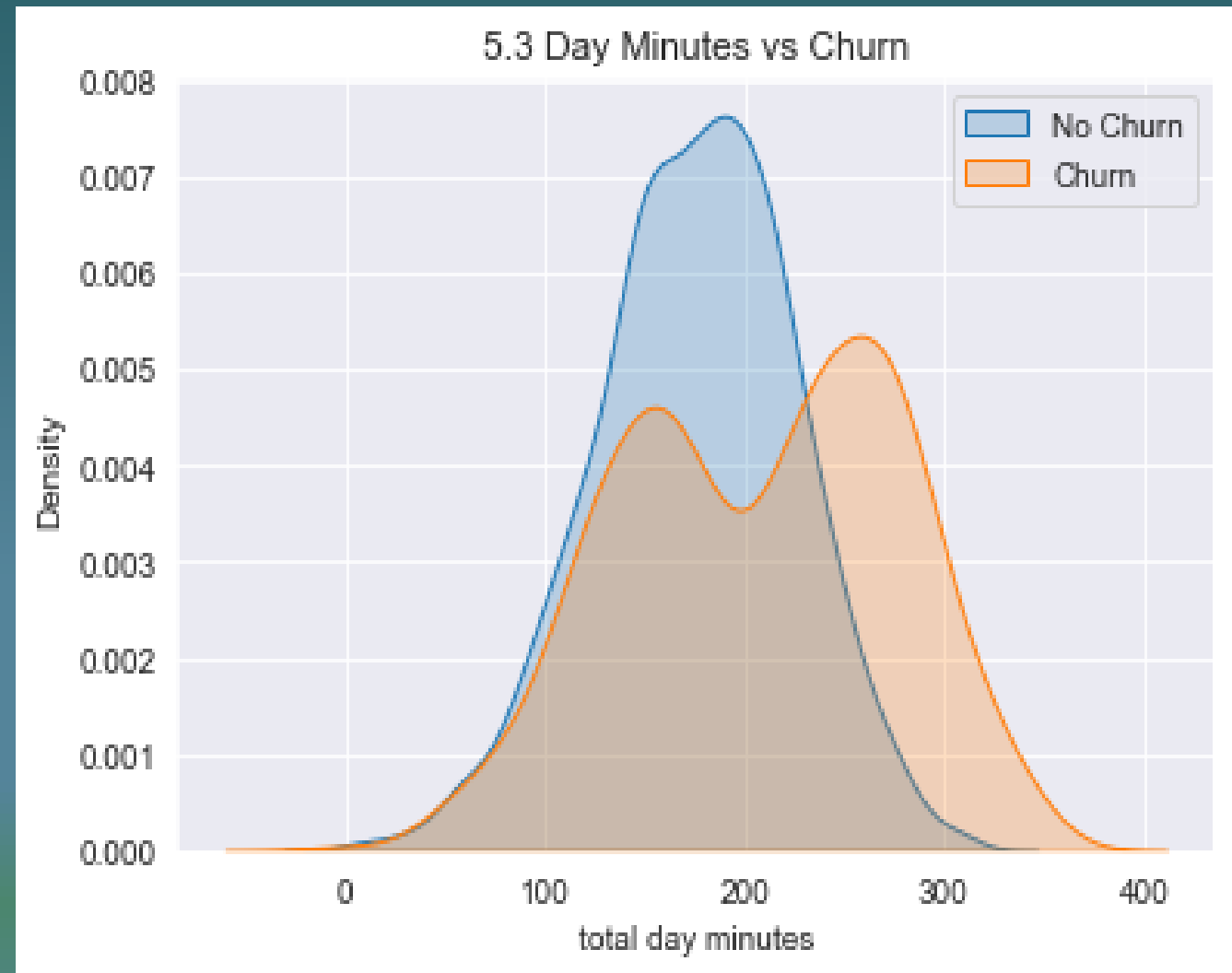
Day Minutes vs Churn (KDE Plot)

Insight

Churned customers tend to have slightly higher usage.

Suggests: -Higher usage → higher bills → possible dissatisfaction

Usage alone is not a strong separator, but still relevant.



Customer Service Calls vs Churn

Insight

Churned customers have more service calls.

Strong indication of:

- Poor customer experience
- Unresolved issues

This is one of the strongest churn drivers



Correlation Heatmap

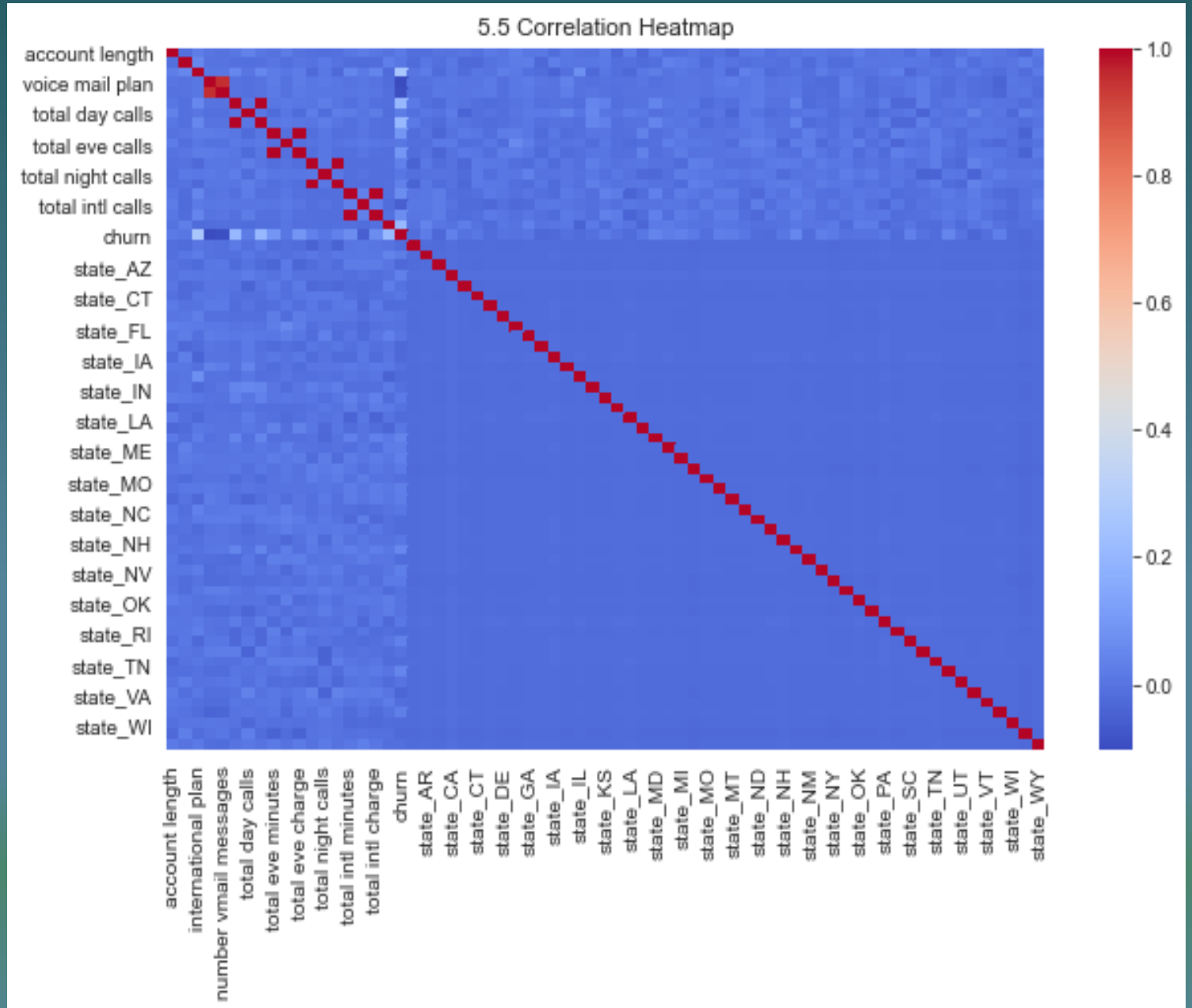
Insight

No extremely strong correlations → no single dominant feature

Moderate relationships exist with:

- Customer service calls
- Charges (day/evening/international)

Indicates churn is influenced by multiple combined factors



Machine Learning Models Built

CONTENT:

- LOGISTIC REGRESSION (BASELINE)
- DECISION TREE
- TUNED DECISION TREE (FINAL MODEL)

We compared multiple models and selected the tuned decision tree based on recall performance.

MODELLING AND EVALUATION

Decision Tree (Better at capturing complex customer behaviors).

Logistic Regression (Simple and interpretable).

- **Interpretation:** Identifying which factors (like high customer service calls) most influence the decision to leave.
- **Visual Suggestion:** A placeholder for your Confusion Matrix (once you run the model).

key insights

- ▶ **Cost-related factors**

(Total day charge, total evening charge, total international charge) → Customers with higher service costs are more likely to churn.

- ▶ **Customer service experience**

(Customer service calls) → A high number of support calls indicates dissatisfaction and increases churn risk.

- ▶ **International service usage**

Customers using international services show higher churn tendency due to higher costs.

- ▶ **High usage patterns**

(Total day minutes, voicemail messages) → Heavy users may churn if pricing is not competitive or value is not perceived as fair.

Model Evaluation (Confusion Matrix)

CONTENT:

- RECALL IS THE MOST IMPORTANT METRIC
- WE FOCUS ON DETECTING CHURNERS CORRECTLY

The confusion matrix shows that the tuned model correctly identifies most churners, reducing missed high-risk customers.

Key Drivers of Customer Churn

CONTENT:

- CUSTOMER SERVICE CALLS
- HIGH CHARGES
- USAGE PATTERNS

The most important factors influencing churn are customer service calls and high usage charges, which can guide business retention strategies.

BUSINESS CONSTRAINTS & RECOMMENDATIONS

- Dataset lacks external competitor pricing and real-time behavioral shifts.
- Actions:
 1. Proactively contact users with >3 service calls.
 2. Review the value proposition of the "International Plan."
 3. Use model probabilities to rank customers by "Risk Score."

conclusion

- ▶ This project successfully developed a machine learning model to predict customer churn in a telecom dataset. After evaluating multiple models, the Tuned Decision Tree was selected as the best-performing model due to its strong recall and balanced performance. The analysis revealed that customer service experience, pricing sensitivity, and international usage patterns are the primary drivers of churn. These insights can help the business design targeted retention strategies to reduce customer attrition and improve long-term profitability.

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